Electric Vehicle Market Performance & Insights

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Dataset Source: ev_market_train.csv
Tools Used: Tableau Public
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1. Introduction

1.1 Project Overview

Electric Vehicle Market Performance & Insights is a data analysis project that explores electric vehicle market data. The project uses an interactive Tableau dashboard to present market information through charts, key metrics, and filters. It helps users explore the data and identify patterns across the years, regions, vehicle categories, or manufacturers included in the dataset.

1.2 Purpose

The purpose of this project is to make electric vehicle market data easier to explore and understand. The dashboard presents the available data in a visual format so users can compare categories and observe trends. The analysis is limited to the fields and time period available in the selected dataset.

2. Ideation Phase

2.1 Problem Statement

PS-1: I am an EV market researcher trying to understand how electric vehicle adoption changes over time. However, it can be difficult to identify important trends by reviewing raw records. This makes it harder to interpret market performance quickly.

PS-2: I am a person interested in the EV market trying to compare the available regions and vehicle categories. However, the information may be difficult to compare when it is not presented in one clear view. This can make it harder to draw conclusions from the data.

2.2 Empathy Map Canvas

Says: “I want to understand how the EV market is changing.” “I need to compare the categories available in the data.”

Thinks: “Which trends are most noticeable?” “Can I compare the data clearly?”

Does: Searches for EV market information and uses charts or filters to explore the available data.

Feels: May feel overwhelmed when reviewing many records, but more confident when the information is presented clearly.

Pains: Raw data can take time to review. Comparing trends and categories may be difficult without visual summaries.

Needs / Gains: A clear overview of the available EV market data, with charts and filters that make comparisons easier.

2.3 Brainstorming

The project ideas considered were:

- Show the year-wise trend in EV market data.
- Compare EV data across the regions included in the dataset.
- Show the distribution of the vehicle categories included in the dataset.
- Compare manufacturers, if manufacturer data is available.

The selected ideas were based on their relevance to the project and whether the required data was available.

3. Requirement Analysis

3.1 Customer Journey Map

The user begins with a question about EV market performance. The user opens the Tableau dashboard and reviews the overview. The user then selects available filters and compares charts or categories. Finally, the user reviews the visual results and uses them to understand patterns in the dataset.

3.2 Solution Requirements

Functional requirements

- The solution should display the EV data available in the selected dataset.
- The dashboard should present relevant metrics and visualizations.
- Users should be able to explore the data using the filters included in the dashboard.
- Charts should update appropriately when a filter is selected.
- The solution should show the selected dataset’s time period and categories clearly.

Non-functional requirements

- The dashboard should be easy to read and navigate.
- Labels, titles, and chart formats should be clear.
- The dashboard should display results consistently.
- The project should identify its dataset source and scope.

3.3 Data Flow Diagram

Insert the **EV Project DFD diagram** here. It shows how data flows from the EV dataset through data preparation and Tableau to the dashboard user. Adjust the diagram labels if your actual workflow differs.

3.4 Technology Stack

Component	Technology Used	Purpose
Data source	Ev_market_train.csv	Provides the EV market data for analysis
Data preparation	No separate preparation tool used	Used to review or prepare the data, if applicable
Data visualization	Tableau Public	Used to create charts and the interactive dashboard
Project documentation	Microsoft Word	Used to prepare the project report

4. Project Design

4.1 Problem–Solution Fit

The project addresses the difficulty of exploring EV market information in raw data form. The Tableau dashboard presents the available information visually and allows users to compare the categories represented in the dataset. This makes the data easier to review and interpret.

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard for exploring the selected EV dataset. It includes visualizations for the fields available in the data and provides filters where applicable. The dashboard is intended to help users review key figures and identify patterns in the dataset.

4.3 Solution Architecture

The project workflow is:

ev_market_train.csv → Tableau data source/workbook → Dashboard visualizations, and filters → EV Market Analyst Tableau dashboard findings → Tableau Story → EV Market Analyst

The dataset is used as the source for the analysis. Tableau connects to the data and displays the resulting visualizations in a dashboard.

5. Project Planning & Scheduling

5.1 Project Planning

Task	Description	Status
Review the dataset	Identify the source, fields, and time period	Completed
Prepare the data	Check and prepare the data using	Not applicable
Define analysis questions	Select questions that can be answered using the available data	Completed
Create visualizations	Build charts in Tableau	Completed
Build the dashboard	Arrange charts and add available filters	Completed
Review the results	Check that the dashboard displays the data as expected	In progress, or Completed after you verify them
Prepare the report	Add documentation, screenshots, and project links	Completed

6. Functional and Performance Testing

6.1 Performance Testing

The dashboard was reviewed to confirm that its charts display the selected EV data and that the available filters work as expected. The visualizations were checked against the fields and values in the dataset.

Checks performed:

- Dataset fields used in the dashboard were reviewed.
- Dashboard charts were checked for appropriate titles and labels.
- Available filters were tested
- Chart values were compared with the source data

7. Results

7.1 Output Screenshots

Figure 1: Tableau Dashboard Overview

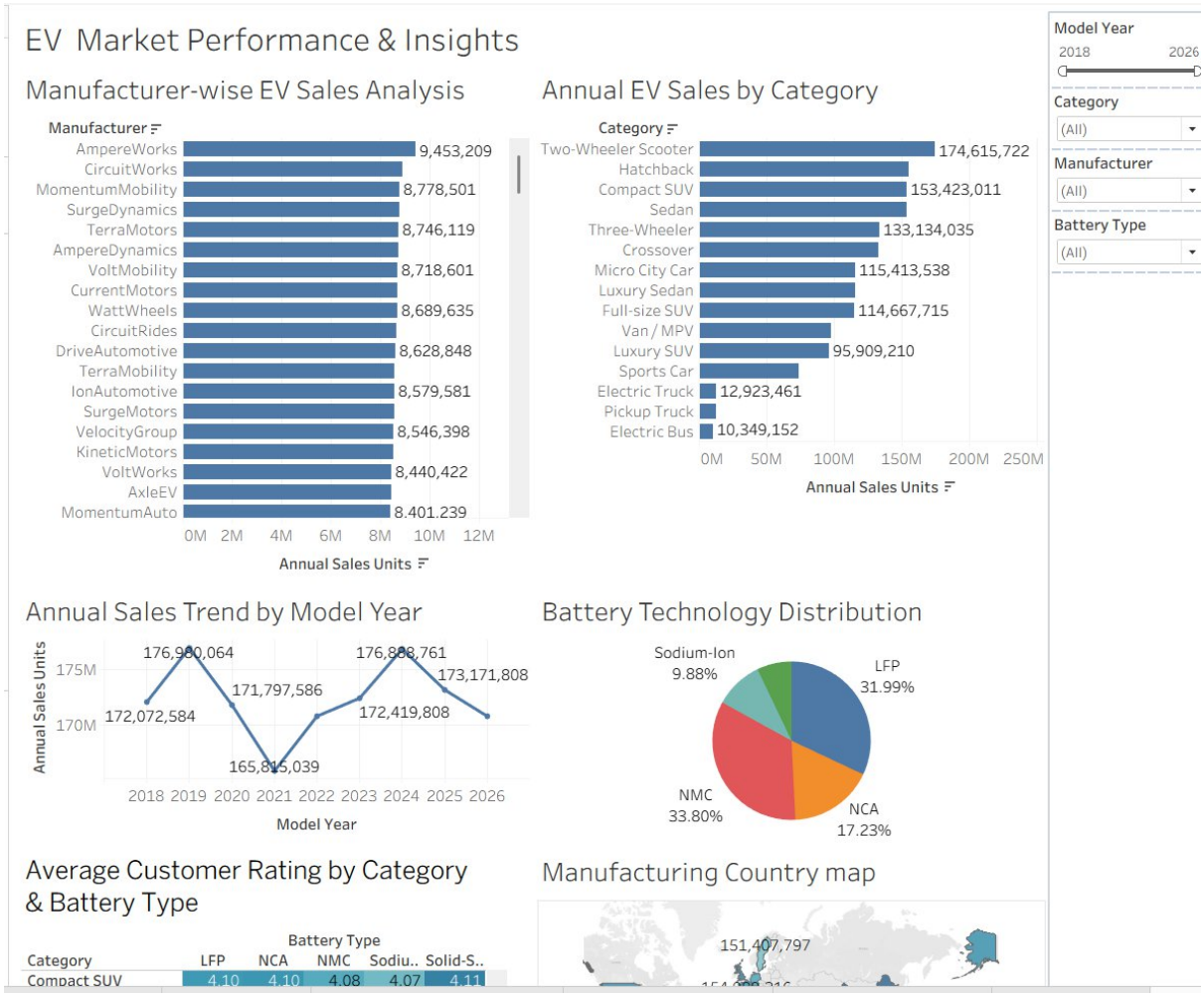


Figure 2: Annual Sales Trend

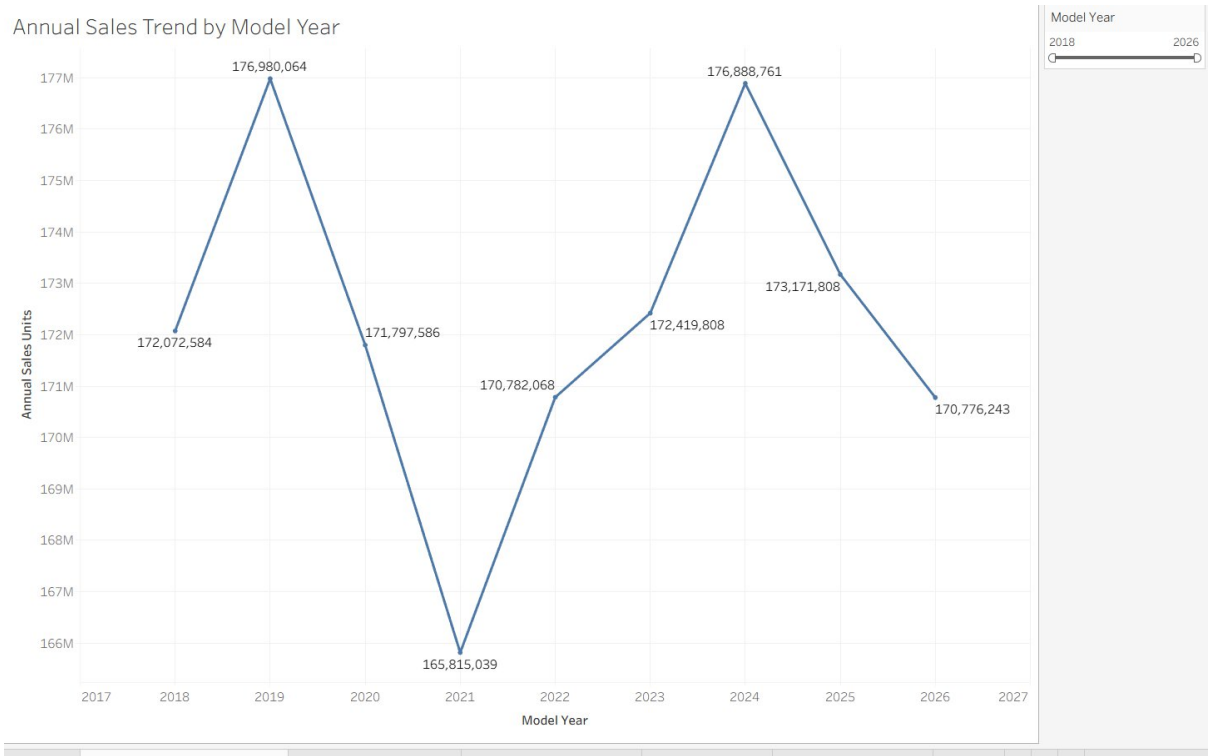
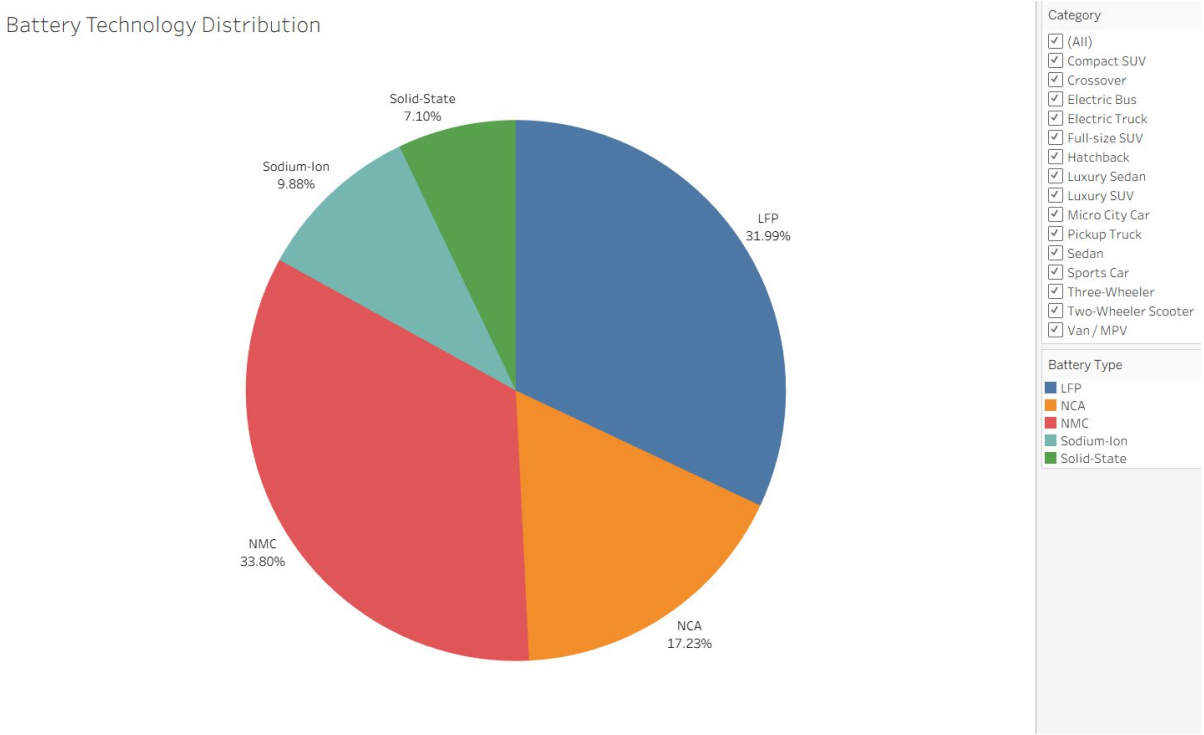


Figure 3: Battery Type Share



8. Advantages & Disadvantages

Advantages

- The dashboard presents the selected EV data visually.
- Charts make it easier to compare the categories included in the dataset.
- Available filters help users explore selected parts of the data.
- The dashboard brings multiple visualizations together in one place.

Limitations

- The analysis is limited to the information available in the selected dataset.
- The results may not represent the entire EV market if the dataset covers only certain years, locations, or categories.
- The dashboard reflects the dataset used when the project was created and may not include live updates.

9. Conclusion

This project uses Tableau to explore the electric vehicle market data available in the selected dataset. The dashboard presents the data through visualizations and, where available, interactive filters. It provides a convenient way to review the included categories and identify patterns. The findings should be interpreted within the scope and limitations of the dataset.

10. Future Scope

The project could be extended by adding newer data, additional regions, or more EV-related measures if reliable data becomes available. The dashboard could also include additional comparisons and updated filters. These are possible future improvements and are not part of the current project unless they have already been implemented.

11. Appendix

- **Dataset source:** ev_market_train.csv (provided dataset)
- **GitHub repository:** manasjain0509/EV-Market_Performance_Insights
- **Tableau Public dashboard:** https://public.tableau.com/app/profile/manas.jain4659/viz/EVMarketPerformanceandInsights/Story1?publish=yes
- **Tableau Public story:** published